

SAFE TRACE

SOP

1. PURPOSE

Automated accident detection and emergency dispatch system.

2. HARDWARE REQUIRED

- ESP32 microcontroller
- ADXL345 Accelerometer
- Neo-6M GPS Module
- Buzzer + LED

3. SOFTWARE REQUIRED

- Libraries: flask, streamlit, pandas, twilio, python-dotenv, requests

4. INSTALLATION

pip install flask ,
Pip install streamlit
pandas twilio python-dotenv requests

5. CONFIGURATION

- Create a .env file with:
TWILION_ACC_SID=your_twilio_sid
TWILION_AUTH_TOKEN=your_twilio_token
TWILION_PHONE_NUMBER=your_twilio_number

6. RUNNING THE SYSTEM

- Start backend: python backend.py
- Start dashboard: streamlit run app.py
- Flash hardware.py to your ESP32 using Arduino IDE

7. TESTING (without hardware)

- Open the dashboard
- Set the G-Force slider to 10.0+ to simulate an accident